

Banafsheh KARIMIAN

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🎓 EDUCATION

COMPUTER ENGINEERING, ARTIFICIAL INTELLIGENCE AND ROBOTICS

OCT. 2020 - PRESENT

University of Tehran

- Supervisors : Dr. Nili (primary), Dr. Hosseini, Dr. Shariatpanahi
- Thesis : Reinforcement Learning from Heterogeneous Demonstrations
- GPA : 19.02/20 – 4.00/4.00 (Via 26 passed credits)
- Ranked 39th - among top 1% in the masters National Universities Entrance Exam

COMPUTER ENGINEERING

SEP. 2015 - JULY. 2019

Iran University of Science and Technology

- GPA : 17.67/20.00 - 3.89/4.00 (Via 141 passed credits)
- Ranked among top 5 students

★ RESEARCH INTERESTS

- Reinforcement Learning
- Robot Learning
- Autonomous Cars
- Machine Learning
- Computer Vision

📖 PUBLICATIONS

1. P. Golshanrad, H. Rahmani, B. Karimian, F. Karimkhani, and G. Weiss, "MEGA : Predicting the best classifier combination using meta-learning and a genetic algorithm," *Intell. Data Anal.*, vol. 25, no. 6, pp. 1547–1563, 2021. [link](#)
2. B. Karimian, A. Mesbah*, E. Mirzaei*, R.Hosseini, P. Shariatpanahi, M. Nili, "Who to Learn from : A Preference-based Method for Social Reinforcement Learning", Ready to be submitted (* equal contributor)

📖 RESEARCH EXPERIENCE

2021-Present

Research Assistant, AT COGNITIVE-SYSTEMS LABORATORY, University of Tehran

- Working on improving RL from heterogeneous demonstrations for robotics applications , [Link](#)
- Worked on Social Reinforcement Learning (Using information observed from other agents in the environment) in order to enhance Reinforcement Learning , [Link](#)
- I also assist two B.Sc. students in extending our previous work on social reinforcement learning to MDP case

Reinforcement Learning Learning from Demonstrations Social Reinforcement Learning Robotics applications

2017 - 2020

Research Assistant, AT DATA-FLOW LABORATORY, Iran University of Science and Technology

- Worked together with a M.Sc. student on her thesis about using meta-learning for predicting the best ensemble of classifiers for a given data set , [Link](#)

Meta Learning Ensemble Methods Classification Methods

📖 TEACHING EXPERIENCE

Present

Teaching Assistant , DISTRIBUTED OPTIMIZATION COURSE, University of Tehran

- Preparing home works on Distributed RL

2021 - Present	Teaching Assistant , INTERACTIVE LEARNING (RL) COURSE, University of Tehran › Designed home works › Mentored students through different phases of the final project
2021 & 2022	Teaching Assistant , MACHINE LEARNING COURSE, University of Tehran › Designed home works › Designed final project and mentored students through different phases of the project
2017 - 2019	Teaching Assistant , DEPARTMENT OF COMPUTER, Iran University of Science and Technology Computer Vision, Algorithm Design and Analysis, Theory of computation, Data Structures, fundamentals of programming › For above courses, I held weekly problem solving sessions and designed home works

</> PROJECTS

ROBOT LOCALIZATION https://banafshehkarimian.github.io/projects/Robot_Localization/ In this project, we used Particle Filter algorithm to localize our Vector robot both in simulation and real world maze. Robot Localization Particle Filters Python ROS Vector Robot	2022
PYBULLET CUSTOMIZED ENVIRONMENTS https://banafshehkarimian.github.io/projects/PyBulletEnv/ An Environment with Franka Panda arm as the robot and multiple Goals for the agent to reach created using PyBullet. PyBullet SAC	2022
GOSSIP TRAINING BEHAVIORAL CLONING FOR SELF DRIVING CARS https://banafshehkarimian.github.io/projects/gossip/ We used gossip training in behavioral cloning for autonomous driving car Python Deep learning Behavioral Cloning	2022
MULTI AGENT FIRE RETARDANT github.com/BanafshehKarimian/forest_fire We used gossip training in behavioral cloning for autonomous driving car Python Deep Learning Reinforcement Learning	2021
More Projects	

≡ SELECTED COURSES

- › Interactive Learning (Reinforcement Learning) assignments
- › Machine Learning
- › Deep learning with application in machine vision and audio processing
- › Artificial Intelligence and Expert Systems
- › Foundation of Computer Vision
- › Advanced Robotics
- › Foundations of Computational Intelligence
- › Distributed Optimization

≡ TECHNICAL SKILLS

Programming Languages	Java, Python, C++.
Other Language	ROS, PyBullet, Webots, OpenCV, Arduino, Pytorch, Keras.
	Native in Persian, Fluent in English : IELTS 7.5 (2018- 2020), GRE 320 (Quant 167, Verbal 153, Writing 3)

“ REFERENCES

* Available upon request